

# Rishi Raval

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## EDUCATION

### Adani University

Bachelor of Technology in Computer Science (AI-ML)

Ahmedabad, Gujarat

Aug 2023 – May 2027

- **CGPA: 8.69/10.0**
- **Core Coursework:** Machine Learning, Artificial Intelligence, Operating Systems, Theory of Computation, Data Structures & Algorithms, Computer Networks, Network Security, Database Management Systems.

### American School of Baroda

Higher Secondary Education (Class XII & X)

Baroda, Gujarat

Graduated May 2023

- **Class XII:** 82.6% | **Class X:** 94.0%

## TECHNICAL SKILLS

**Languages:** C++, C, Python, Java, JavaScript, SQL    **Web:** React, HTML/CSS

**Quant & ML:** NumPy, Pandas, Scikit-learn, Matplotlib

**Tools/OS:** Git, Linux (Fedora, Kali, Arch), VS Code, Jupyter Notebooks

## PROJECTS

### Multimodal Direction & Volatility Forecasting | *PyTorch, FinBERT, GAT, CNN-BiLSTM*

[\[GitHub\]](#)

- Engineered a **gated attention fusion model** that integrates price time-series (CNN-BiLSTM), SEC filings (FinBERT), and news sentiment to forecast stock behavior[cite: 12, 312].
- Developed a custom **Graph Attention Network (GAT)** in pure PyTorch to model inter-company signals (suppliers/competitors), yielding a **+2.3% AUC improvement** over baseline[cite: 212, 373].
- Achieved a **volatility prediction  $R^2$  of 0.41** and a selective direction **AUC of 0.60** through confidence-based prediction filtering at 6.3% coverage[cite: 14, 578].

### FastGreeks — High-Performance Quant Library | *C++, Templates, Math*

[\[GitHub\]](#)

- Developed a **header-only C++ library** for real-time calculation of Option Greeks (Delta, Gamma, Vega, Theta, Rho) using the Black-Scholes-Merton model.
- Implemented an **Implied Volatility solver** using the Newton-Raphson method, optimized for numerical stability in edge-case market conditions.

### AlphaVision — Stock Price Direction Predictor | *XGBoost, Scikit-learn, Ensemble Learning*

[\[GitHub\]](#)

- Engineered a cross-ticker **XGBoost classifier** predicting intraday movement with **60.2% accuracy** on unseen data, significantly outperforming the random baseline.
- Evaluated 28 configurations across 4 algorithms (Regression, Classification, Ensembles); final model utilized 76 normalized technical indicators for cross-stock generalization.

### Quantitative Backtesting Engine | *Python, Pandas, Vectorized Backtesting*

[\[GitHub\]](#)

- Built a vectorized framework to evaluate **mean-reversion and momentum strategies** on historical equity data, optimizing for execution speed and data integrity.
- Computed high-fidelity risk metrics including **Sharpe Ratio**, **Sortino Ratio**, and **Max Drawdown** to quantify strategy alpha and robustness.

## CERTIFICATIONS & ACHIEVEMENTS

### JPMorgan Chase & Co. – Quantitative Research Virtual Experience | *Forage*

Nov 2025

- Simulated the work of a Quantitative Researcher; analyzed loan books to estimate default probability.
- Utilized dynamic programming to bucket FICO scores and visualize credit risk data.

### Competitive Programming | *LeetCode*

- Solved **550+ problems** across Easy, Medium, and Hard difficulties.
- Contest Rating: **1444**

### Career Essentials in Generative AI | *Microsoft & LinkedIn*

Jul 2024

- Acquired foundational skills in Generative AI and Prompt Engineering.

## INTERESTS

**Quant Finance:** HFT Systems, Market Microstructure    **Writing:** [Recursify](#) & [Rishi Writes](#)    **Hobbies:** Linux, Photography